

Notice ID Number NB681010-25-00919

NOTICE OF INTENT TO SOLE SOURCE

Title: Sources Sought Notice for Widely Tunable Near Infrared Laser

This is a notice of intent the Acquisition Management Division (AMD) at the National Institute of Standards and Technology Physical Measurements Lab, Photonics and Optomechanics Group intends to procure a widely tunable near infrared laser on a sole source basis with Toptica Photonics, Inc using Simplified Acquisition Procedures for Commercial Items (FAR Parts 12 and 13).

This notice shall not be construed as a Request for Quote. Respondents are advised that the U.S. Government will not pay for information or administrative costs incurred in the response to this notice. All costs associated with responding to the notice will be solely at the interested party's expense. Not responding to this notice does not preclude participation in any future RFQ or other solicitation (if any are issued). This is a notice of intent to sole source, not a pre-solicitation notice. There is no formal solicitation available at this time. No award will be made as a result of this notice. Respondents will not be notified of the results of this notice, or results of information submitted.

The Photonics and Optomechanics Group is developing next-generation optical technologies that require laser readout methods at a wide variety of wavelengths, including (but not limited to) Cesium electronic transitions near 885 and 895 nanometers and quantum dot transitions between 900 and 940 nanometers. Hence, a laser that operates at these wavelengths, as well as surrounding wavelengths with as large a tuning range as possible, is needed to carry out these experiments. It was determined by extensive market research that Toptica Photonics is the only vendor currently offering a product (CTL 900) that meets all the technical specifications.

The success of experiments (i.e., the number of devices we can test) is directly proportional to this wavelength tuning range. No other commercial vendors offer a mode-hop-free laser package that covers the specific Cesium and quantum dot wavelengths we plan to interrogate.

TECHNICAL REQUIREMENTS

Technical Specifications:

1. The laser wavelength must be continuously tunable, without mode hops, from 880 nm to 930 nm.
2. The delivered product must include the necessary control electronics to set and scan/modulate the laser diode current and piezo voltage; the latter must enable >50 GHz laser frequency scans, without mode hops.
3. The laser shall deliver >10 mW (with optical isolation) into a single-mode optical fiber across its wavelength range.
4. It shall exhibit a short-term linewidth <1 kHz and integral linewidth <1 MHz.

5. The controller must have an interactive touch screen for selecting operational modes, setting parameters, and visualizing signals at provided input ports.
6. The laser wavelength must be tunable using a coarse motion stage, a piezoelectric motion stage, and the laser diode current. These tuning methods must be able to operate simultaneously.
7. Mode-hop free wavelength range: 880-930 nanometers
8. Nominal fiber-coupled power: at least 10 mW across wavelength range
9. Typical instantaneous linewidth: < 1 kHz
10. Typical integral linewidth: < 1 MHz
11. Typical short-term linewidth (5 us integration): < 10 kHz
12. Beam polarization: Linear > 100:1
13. Fiber coupling efficiency: > 50%
14. Optical isolation: > 50 dB
15. Piezo optical frequency scan range: > 50 GHz
16. Maximum wavelength scan speed: > 8 nm/s
17. Piezo modulation bandwidth: > 100 Hz
18. Diode current port modulation bandwidth: > 100 kHz

NOTICE RESPONSE

Interested firms/companies are encouraged to respond to this notice by submitting documentation that clearly demonstrates how the minimum requirements and technical specifications will be met. Response which merely repeat the minimum requirements, and technical specifications without providing demonstration of how the minimum requirements and technical specifications will be met, will not be considered or evaluated by the government. Responses can demonstrate their capabilities to meet the requirements and specifications described above by providing brochures, manuals, reports, demonstration videos, other technical data, and identification of current customers.

In responding to this notice, please DO NOT PROVIDE PROPRIETARY INFORMATION. In addition to the technical documentation, the response must include the following:

- Company name and address
- Point of Contact for questions and/or clarifications, including telephone number and email address
- Teaming partners (if applicable)
- For supplies – If the supply is a commercial off the shelf item, modified commercial off the shelf item, or prototype.
- Any additional information as deemed pertinent by the respondent.

Interested responses must be submitted to Contract Specialist Ranae Armstrong at ranae.armstrong@nist.gov with a CC to Contracting Officer Don Collie at Donald.collie@nist.gov. no later than 04/17/25, 12:00pm, ET. Telephone and faxed responses will not be accepted.